A Growth Bound for Arbitrarily Interleaved Transfer Bridges

Collatz proof project — September 5, 2026

Let T be the shortcut Collatz map and $B(x) = 9x+2$. Consider a finite path starting at a positive integer n, using ordinary T steps and bridges B in any order. Let x be its endpoint, t its number of ordinary steps, j its number of odd ordinary steps, and b its number of bridges. Set $e = 2j + 4b$. We impose no congruence or recursive-parameter restrictions on bridges. This relaxation makes the exclusion below apply to every admissible forward transfer-bridge path as well.

Theorem 1 (Mixed-path growth bound)**.** *Every such path satisfies*

$$2^t x \le 2^e n.$$

Proof. An even step divides the current value by two. At a positive odd value y, $2T(y) = 3y+1 \le 4y$, so an odd step increases size by at most a factor of two. A bridge satisfies $9y + 2 \le 16y$ for $y \ge 1$. Multiplying these bounds gives $x \le 2^{j-(t-j)+4b} n = 2^{e-t} n$. The Lean proof uses natural-number inequalities and induction on the path, without negative powers, logarithms, or a convergence premise. □

Theorem 2 (Cycle charge cannot decrease)**.** *If the starting and ending values both lie in* $\{1, 2\}$, *then*

$$t + \mathbf{1}_{n=1} \le e + \mathbf{1}_{x=1}.$$

Equivalently, a symbolic path with initial slope exponent p *and initial clock* s *cannot lower*

$$L(y, p, s) = 2p - s + \mathbf{1}_{y=1}$$

between its starting and ending cycle states.

Proof. Write $n = 2^{1-\mathbf{1}_{n=1}}$ and $x = 2^{1-\mathbf{1}_{x=1}}$ in the growth bound and compare powers of two. The symbolic slope exponent increases by $j + 2b = e/2$ and the clock by t, giving the equivalent charge comparison. Lean checks the natural-number inequality by the four possible cycle endpoint cases. □

Consequence for transfer search. At clock s, suppose a source and target have cycle base values and their symbolic endpoints are $x + 3^p 2^{D-s} Q$ and $y + 3^q 2^{D-s} Q$. If $L(x, p, s) > L(y, q, s)$, no subsequent forward source path, using any number of bridges, can match the target's ordinary continuation in both constant and slope. The target remains in the cycle and preserves its charge. At a fixed clock, equal symbolic endpoints require equal cycle state and slope exponent, hence equal charge, contradicting the theorem. This allows sound pruning without imposing a future bridge-count bound.

What changed from the earlier cycle experiment. Previously, the exclusion covered repeated $2 \mapsto 20$ excursions with no bridges inside an excursion. The new theorem allows bridges anywhere along a forward path, including inside excursions. The earlier finite sample's 302 negative-gap base pairs therefore cannot be rescued by any forward bridge path started only after both unmodified base orbits have entered the cycle. Earlier auxiliary paths, inverse steps, and other inference rules are outside this conclusion.

Verification and remaining goal. The module `Collatz.BridgeGrowth` proves positivity, the growth bound, and the cycle inequality. The search now removes the excluded states. A complete comparison on parameters 0–1000 preserves certificate outcomes and first-hit depths with pruning enabled or disabled; all eight affine search tests pass. The census remains Python evidence. The selected Lean axiom audit contains 217 declarations, using only standard foundations.

This result limits a proof-search mechanism; it does not establish global transfer coverage or prove Collatz. The next useful extension must reach these cases before the late-cycle obstruction applies or introduce a transition not covered by this forward-path bound. No priority claim is made.